IBP for V++ Massive Bubble

Zhehan Qin[a,b]*

[a] *Department of Physics, Tsinghua University, Beijing 100084, China*

[b] *Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Wilberforce Road, Cambridge, CB3 0WA, UK*

Abstract

In this note we construct IBP relations for the one-loop two-point (bubble) diagram in de Sitter space with both vertices being plus (V++) and internal propagators massive. We follow the structure of the reference note for V+- massive bubble, adding only the modifications required by the θ-function time-ordering. The momentum IBP relations remain unchanged, while time IBP relations acquire tadpole terms from derivatives of θ-functions. We provide the complete IBP system ready for Mathematica implementation.

*Email: qzh21@mails.tsinghua.edu.cn

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1 Background, goals, and structure

The goal is to construct practical reduction relations for the massive V++ de Sitter bubble, including the tadpole families induced by time ordering. Differential operators in external kinematics map integrals within the same families; after reduction to a master basis, these operators induce a first-order linear system for the master-integral vector.

We organize the note as follows. Section 2 fixes conventions and defines the $G^{(12)}$, $G^{(21)}$ and tadpole R families. Section 3 lists the IBP system and the kinematic differential operators. Section 4 records Baikov and dimension-shift relations for the momentum integral. Section 5 summarizes a $d\log$/intersection-theory viewpoint tailored to the present special-function setting.

2 Basic definitions, notation, and integral families

Notations The metric reads: $\mathrm{d}s^2 = a^2(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2)$ with scale factor $a(\tau) = -1/(H\tau)$ and $\tau \in (-\infty, 0)$. We set $H \equiv 1$ throughout. We use $\theta_{ij} \equiv \theta(\tau_i - \tau_j)$ and $\theta(0) = 1/2$. We work in D spatial dimensions (so the loop momentum integration is $\int \mathrm{d}^D\mathbf{q}$); in dimensional regularization one may take $D = 3-2\epsilon$. For simplicity we take all mass parameters equal, $\nu_1 = \nu_2 = \nu_3 = \nu_4 \equiv \nu$, and we also specialize to equal external energies $P_1 = P_2$.

Hankel building blocks Following the notation of arXiv:2401.00129v5 (Eqs. 3.11–3.12), we define the rescaled Hankel functions

$$h_\nu^{(1)}(x) \equiv x^{-\nu}\mathrm{H}_\nu^{(1)}(x), \qquad h_\nu^{(2)}(x) \equiv x^{-\nu}\mathrm{H}_{\nu^*}^{(2)}(x), \qquad \nu \in \mathbb{C}, \quad x > 0. \tag{1}$$

They satisfy (Eq. (3.13) in arXiv:2401.00129v5)

$$h''(x) + \frac{2\nu+1}{x} h'(x) + h(x) = 0. \tag{2}$$

The mass parameter is $m^2 = 9/4 - \nu^2$. We also define derivatives

$$h_{\nu,0}(x) \equiv h_\nu(x), \qquad h_{\nu,n}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n} h(x), \tag{3}$$

For the two branches we also write

$$h^{(1)}_{\nu,n}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n} h^{(1)}_\nu(x), \qquad h^{(2)}_{\nu,n}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n} h^{(2)}_\nu(x). \tag{4}$$

When the mass label is clear from context, we write $h_n \equiv h_{\nu,n}$ and similarly $h^{(1,2)}_n \equiv h^{(1,2)}_{\nu,n}$.

V++ bulk-to-bulk propagator and time ordering For a massive scalar on the + branch, the Schwinger–Keldysh propagator contains two time orderings,

$$G^{++}_\nu(k;\tau_1,\tau_2) \;\propto\; h^{(1)}_\nu(-k\tau_1)\, h^{(2)}_\nu(-k\tau_2)\, \theta_{12} \;+\; h^{(2)}_\nu(-k\tau_1)\, h^{(1)}_\nu(-k\tau_2)\, \theta_{21}, \tag{5}$$

with $\tau_1, \tau_2 \in (-\infty, 0)$. For the bubble, the product of two G^{++} propagators naturally splits into two contributions proportional to θ_{12} and θ_{21} respectively. This motivates treating the two orderings as two integral families related by symmetry.

Two integral families To clearly distinguish the two time-orderings, we define two integral families with explicit θ-function labels:

Family with θ_{12}

$$\begin{aligned} G^{(12)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \equiv & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2}\, \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ & \times h^{(1)}_{n_1}(-q\tau_1) h^{(2)}_{n_2}(-q\tau_2) \\ & \times h^{(1)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau_1) h^{(2)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\, \theta_{12}. \end{aligned} \tag{6}$$

Family with θ_{21}

$$\begin{aligned} G^{(21)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \equiv & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2}\, \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ & \times h^{(2)}_{n_1}(-q\tau_1) h^{(1)}_{n_2}(-q\tau_2) \\ & \times h^{(2)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau_1) h^{(1)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\, \theta_{21}. \end{aligned} \tag{7}$$

Relation between the two families The two families are related by exchanging time variables and external energies:

$$G^{(21)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\};P_1,P_2] = G^{(12)}[\{n_2,n_1,n_4,n_3\},\{a_2,a_1\},\{b_1,b_2\};P_2,P_1]. \tag{8}$$

Complete V++ correlator The physical V++ bubble correlator is the sum:

$$\mathcal{V}_{++} = G^{(12)} + G^{(21)}. \tag{9}$$

Tadpole families Time IBP relations involve total derivatives $\int \partial_{\tau_i}[\cdots] = 0$. When the integrand contains θ_{12} or θ_{21}, one generates coincidence terms proportional to $\delta(\tau_1 - \tau_2)$, which reduce the two-time integral to a one-time integral. We treat these coincidence contributions as tadpole families R_1, R_2, which depend only on $P_0 \equiv P_1 + P_2$ (and we keep P_0 implicit in the notation).

$$\begin{aligned} R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \equiv\ & e^{\pi \mathrm{Im}\,[\nu]} \frac{4i}{\pi} \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{|\mathbf{q}|^{b_1} |\mathbf{k}_s + \mathbf{q}|^{b_2}} \\ & \times h^{(1)}_{n_3}(-|\mathbf{q} + \mathbf{k}_s|\tau) h^{(2)}_{n_4}(-|\mathbf{q} + \mathbf{k}_s|\tau), \end{aligned} \tag{10}$$

$$\begin{aligned} R_2[\{n_1, n_2\}, \{a\}, \{b_1, b_2\}] \equiv\ & e^{\pi \mathrm{Im}\,[\nu]} \frac{4i}{\pi} \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{|\mathbf{q}|^{b_1} |\mathbf{k}_s + \mathbf{q}|^{b_2}} \\ & \times h^{(1)}_{n_1}(-q\tau) h^{(2)}_{n_2}(-q\tau). \end{aligned} \tag{11}$$

Direct Tadpole Family According to the Feynman rules, the tadpole diagram is given by the one-time integral of the equal-time propagator. Adopting the regularization $\theta(0) = 1/2$, the equal-time V++ propagator is

$$G^{++}_{\nu}(q; \tau, \tau) = \frac{1}{2}\left[h^{(1)}_{\nu}(-q\tau) h^{(2)}_{\nu}(-q\tau) + h^{(2)}_{\nu}(-q\tau) h^{(1)}_{\nu}(-q\tau)\right] = h^{(1)}_{\nu}(-q\tau) h^{(2)}_{\nu}(-q\tau). \tag{12}$$

We define the direct tadpole family T corresponding to this diagrammatic contribution:

$$\begin{aligned} T[\{n_1, n_2\}, \{a\}, \{b\}] \equiv\ & \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{|\mathbf{q}|^b} \\ & \times h^{(1)}_{n_1}(-q\tau) h^{(2)}_{n_2}(-q\tau). \end{aligned} \tag{13}$$

The families R_1, R_2 instead arise as remaining terms in time IBP for the bubble: one propagator is contracted using the Wronskian, producing the universal prefactor $e^{\pi \mathrm{Im}\,[\nu]} \frac{4i}{\pi}$ and an extra $(-k\tau)^{-2\nu-1}$ factor for the contracted line. In the integral-family language this is implemented as shifts of the τ-power and the corresponding $|k|$-power indices (e.g. $a \to a - 2\nu - 1$ and $b \to b + 2\nu + 1$), together with the constraint $\delta_{n_i + n_j, 1}$ from the contraction formula.

3 Reduction relations for the one-loop two-point bubble

3.1 Top-sector families $G^{(12)}, G^{(21)}$

Origin of the tadpole (remaining) terms Time IBP relations involve total derivatives $\int \partial_{\tau_i}[\cdots] = 0$. With θ_{12} or θ_{21} present, derivatives generate coincidence terms proportional to $\delta(\tau_1 - \tau_2)$. For Hankel-type integrands, the coincidence terms collapse one time integration using the Wronskian of the $h^{(1,2)}_{\nu}$ pair, producing one-time tadpole families. In the notation of arXiv:2401.00129v5 (Section 3.7), in the time derivation of G_{++}, the θ functions yield a remaining term

$$h^{(1)}_{n_1}(-k\tau_i)\, h^{(2)}_{n_2}(-k\tau_i) - h^{(2)}_{n_1}(-k\tau_i)\, h^{(1)}_{n_2}(-k\tau_i) = \delta_{n_1+n_2,1} e^{\pi \mathrm{Im}[\nu]} \frac{4i}{\pi} (-k\tau_i)^{-2\nu-1}\,. \tag{14}$$

which, in our integral-family language, is implemented as a shift of the corresponding $|k|$-power index by $2\nu+1$ and the τ-power index by $-2\nu-1$, together with the contraction of the two-time integrations by one integral in the remaining family.

In particular, since the R families originate from contracting one propagator in the top sector, the contracted line carries an extra $(-k)^{-2\nu-1}$ factor from the Wronskian. In the Mathematica/Kira implementation we keep the momentum baselines trivial,

$$|q|^{-b_1}\,|\boldsymbol{k}_s+\boldsymbol{q}|^{-b_2}, \qquad b_{10}=0, \quad b_{20}=-2\nu,$$

and the remaining-term factor is tracked explicitly by the b shifts (e.g. $b_1 \to b_1+2\nu+1$) in Eqs. (15)–(16).

Time IBP with θ-function tadpole terms In the following, G denotes $G^{(12)}+G^{(21)}$. The combination of the two parts can employ the propagator contraction formula, thereby simplifying the calculation. Momentum IBP and equation-of-motion relations apply to the families first. The only new ingredient in V++ (compare to V+-) is the θ function contributions in the time IBP.

For ∂_{τ_1}:

$$\begin{aligned}0 =& -\mathrm{i}P_1 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_1 G[\{n_1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1,b_2\}]\\ &- G[\{n_1+1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - G[\{n_1,n_2,n_3+1,n_4\},\{a_1,a_2\},\{b_1,b_2-1\}]\\ &+ \delta_{n_1+n_2,1}(-1)^{n_1+1}R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}]\\ &+ \delta_{n_3+n_4,1}(-1)^{n_3+1}R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}],\end{aligned} \tag{15}$$

and for ∂_{τ_2}:

$$\begin{aligned}0 =& -\mathrm{i}P_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2-1\},\{b_1,b_2\}]\\ &- G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2\},\{b_1,b_2-1\}]\\ &+ \delta_{n_1+n_2,1}(-1)^{n_2+1}R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}]\\ &+ \delta_{n_3+n_4,1}(-1)^{n_4+1}R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}],\end{aligned} \tag{16}$$

where ν is the common mass parameter.

The tadpole families R_1, R_2 are defined in Section 2.

Momentum IBP Momentum IBP relations are unchanged compared to the V+− bubble, because $\partial_{\boldsymbol{q}^i}\theta_{ij}=0$. The derivation can be organized in the same chain-rule form as in Section 1.6 of the reference note by introducing

$$x_1 \equiv |\boldsymbol{q}|, \qquad x_2 \equiv |\boldsymbol{q}+\boldsymbol{k}_s|, \qquad \mathcal{O}_1 \equiv \boldsymbol{q}^i\partial_{\boldsymbol{q}^i}, \qquad \mathcal{O}_2 \equiv (\boldsymbol{q}^i+\boldsymbol{k}_s^i)\partial_{\boldsymbol{q}^i}, \tag{17}$$

whose actions on (x_1,x_2) are

$$\begin{aligned}\mathcal{O}_1 x_1 &= x_1,\\ \mathcal{O}_1 x_2 &= \frac{1}{2}x_2^{-1}(x_1^2+x_2^2-k_s^2),\\ \mathcal{O}_2 x_1 &= \frac{1}{2}x_1^{-1}(x_1^2+x_2^2-k_s^2),\\ \mathcal{O}_2 x_2 &= x_2.\end{aligned} \tag{18}$$

Equivalently, for any function $F(x_1, x_2)$ one has the chain rule

$$\mathcal{O}_\alpha F = (\mathcal{O}_\alpha x_1)\,\partial_{x_1} F + (\mathcal{O}_\alpha x_2)\,\partial_{x_2} F, \qquad \alpha = 1, 2. \tag{19}$$

From $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ one may write an intermediate step in terms of the composite operator acting on the full integrand,

$$\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_1\Big] \\
&\Bigg[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h^{(1)}_{n_1}(-x_1\tau_1) h^{(2)}_{n_2}(-x_1\tau_2) \times h^{(1)}_{n_3}(-x_2\tau_1) h^{(2)}_{n_4}(-x_2\tau_2)\Bigg] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_1 x_1)\partial_{x_1} + (\mathcal{O}_1 x_2)\partial_{x_2}\Big] \\
&\Bigg[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h^{(1)}_{n_1}(-x_1\tau_1) h^{(2)}_{n_2}(-x_1\tau_2) \times h^{(1)}_{n_3}(-x_2\tau_1) h^{(2)}_{n_4}(-x_2\tau_2)\Bigg].
\end{aligned} \tag{20}$$

Expanding the derivatives using $\partial_x x^{-b} = -(b/x)x^{-b}$ and $\partial_x h_n(-x\tau) = (-\tau)\, h_{n+1}(-x\tau)$ yields the closed relation in terms of shifted G integrals:

$$\begin{aligned}
0 = {} & D\, G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& - b_1\, G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_2 \Big(G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& \qquad + G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1 - 2, b_2 + 2\}] \\
& \qquad - k_s^2\, G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2 + 2\}] \Big) \\
& + G[\{n_1 + 1, n_2, n_3, n_4\}, \{a_1 + 1, a_2\}, \{b_1 - 1, b_2\}] \\
& + G[\{n_1, n_2 + 1, n_3, n_4\}, \{a_1, a_2 + 1\}, \{b_1 - 1, b_2\}] \\
& + \frac{1}{2} \Big(G[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2 - 1\}] \\
& \qquad + G[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1 + 2, b_2 + 1\}] \\
& \qquad - k_s^2\, G[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2 + 1\}] \Big) \\
& + \frac{1}{2} \Big(G[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1, b_2 - 1\}] \\
& \qquad + G[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2\, G[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1, b_2 + 1\}] \Big),
\end{aligned} \tag{21}$$

where D is the spatial dimension (in dimensional regularization $D = 3 - 2\epsilon$). The second momentum IBP can be written in the same chain-rule form as

$$\begin{aligned}
0 = {} & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_2\Big] \\
& \Bigg[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h^{(1)}_{n_1}(-x_1\tau_1) h^{(2)}_{n_2}(-x_1\tau_2) \times h^{(1)}_{n_3}(-x_2\tau_1) h^{(2)}_{n_4}(-x_2\tau_2)\Bigg],
\end{aligned} \tag{22}$$

and, after expansion, it gives the symmetric relation with the exchanges

$$\{n_1, n_2, n_3, n_4\} \leftrightarrow \{n_3, n_4, n_1, n_2\},$$
$$\{b_1, b_2\} \leftrightarrow \{b_2, b_1\}.$$

Equation of motion relations The Bessel equation (2) yields:

$$\begin{aligned}
G[\{2, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- G[\{0, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
&- (2\nu + 1) G[\{1, n_2, n_3, n_4\}, \{a_1 - 1, a_2\}, \{b_1 + 1, b_2\}], \quad (23) \\
G[\{n_1, 2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- G[\{n_1, 0, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
&- (2\nu + 1) G[\{n_1, 1, n_3, n_4\}, \{a_1, a_2 - 1\}, \{b_1 + 1, b_2\}], \quad (24) \\
G[\{n_1, n_2, 2, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- G[\{n_1, n_2, 0, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
&- (2\nu + 1) G[\{n_1, n_2, 1, n_4\}, \{a_1 - 1, a_2\}, \{b_1, b_2 + 1\}], \quad (25) \\
G[\{n_1, n_2, n_3, 2\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- G[\{n_1, n_2, n_3, 0\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
&- (2\nu + 1) G[\{n_1, n_2, n_3, 1\}, \{a_1, a_2 - 1\}, \{b_1, b_2 + 1\}], \quad (26)
\end{aligned}$$

where ν is the common mass parameter.

Symmetry relations 1. **Propagator exchange**:

$$G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = G[\{n_3, n_4, n_1, n_2\}, \{a_1, a_2\}, \{b_2, b_1\}]. \quad (27)$$

2. **Time exchange with energy swap**:

$$G^{(21)}[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = G^{(12)}[\{n_2, n_1, n_4, n_3\}, \{a_2, a_1\}, \{b_1, b_2\}; P_2, P_1]. \quad (28)$$

With the specialization $P_1 = P_2$, this becomes a direct symmetry relation between the two time-orderings at fixed kinematics:

$$G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_1] = G[\{n_2, n_1, n_4, n_3\}, \{a_2, a_1\}, \{b_1, b_2\}; P_1, P_1]. \quad (29)$$

Differential operators in kinematics The dependence on external energies is generated by derivatives with respect to P_1, P_2:

$$\frac{\partial}{\partial P_1} G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = i\, G[\{n_1, n_2, n_3, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2\}; P_1, P_2], \quad (30)$$

$$\frac{\partial}{\partial P_2} G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = i\, G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2 + 1\}, \{b_1, b_2\}; P_1, P_2]. \quad (31)$$

Derivative with respect to k_s We use

$$\frac{\partial}{\partial k_s}|\boldsymbol{q}+\boldsymbol{k}_s| = \frac{k_s^2+\boldsymbol{k}_s\cdot\boldsymbol{q}}{k_s|\boldsymbol{q}+\boldsymbol{k}_s|} = \frac{k_s^2+|\boldsymbol{q}+\boldsymbol{k}_s|^2-q^2}{2k_s|\boldsymbol{q}+\boldsymbol{k}_s|}, \tag{32}$$

which leads to a closed relation within the same G family:

$$\begin{aligned}
&\frac{\partial}{\partial k_s}G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}]\\
&= -\frac{b_2}{2k_s}\Big(k_s^2\,G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}]\\
&\qquad + G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}]\\
&\qquad - G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}]\Big)\\
&+\frac{1}{2k_s}\Big(k_s^2\,G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}]\\
&\qquad + G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}]\\
&\qquad - G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1-2,b_2+1\}]\Big)\\
&+\frac{1}{2k_s}\Big(k_s^2\,G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}]\\
&\qquad + G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}]\\
&\qquad - G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}]\Big).
\end{aligned} \tag{33}$$

Scaling relation:

$$\begin{aligned}
&\left(k_s\frac{\partial}{\partial k_s}+P_1\frac{\partial}{\partial P_1}+P_2\frac{\partial}{\partial P_2}\right)G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}]\\
&= (D-b_1-b_2-a_1-a_2-2)G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}].
\end{aligned} \tag{34}$$

3.2 Tadpole families R_1, R_2

Tadpole families R_1, R_2 The tadpole terms R_1, R_2 are single-time integrals generated by the time IBP coincidence terms. They require their own IBP system for reduction.

Kira check: whether b_1 can be reduced to 0 todo

IBP relations for R_1 (and R_2 via symmetry) The tadpole family R_1 satisfies the following IBP relations. The corresponding relations for R_2 follow via the symmetry $\boldsymbol{q}\to-(\boldsymbol{q}+\boldsymbol{k}_s)$, which exchanges $b_1\leftrightarrow b_2$ and relabels Hankel indices appropriately.

Time IBP for R_1:

$$\begin{aligned}
0 =& -i(P_1+P_2)R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - aR_1[\{n_3,n_4\},\{a-1\},\{b_1,b_2\}]\\
&- R_1[\{n_3+1,n_4\},\{a\},\{b_1,b_2-1\}] - R_1[\{n_3,n_4+1\},\{a\},\{b_1,b_2-1\}].
\end{aligned} \tag{35}$$

Momentum IBP for R_1: from $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$,

$$
\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_1\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_1 x_1)\partial_{x_1} + (\mathcal{O}_1 x_2)\partial_{x_2}\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right]. \qquad (36)
\end{aligned}
$$

$$
\begin{aligned}
0 =& D R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] - b_1 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_2 \Big(R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] + R_1[\{n_3, n_4\}, \{a\}, \{b_1 - 2, b_2 + 2\}] \\
& - k_s^2 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2 + 2\}] \Big) \\
& + \frac{1}{2} \Big(R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2 R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 + 1\}] \Big) \\
& + \frac{1}{2} \Big(R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2 R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 + 1\}] \Big). \qquad (37)
\end{aligned}
$$

Complementary momentum IBP for R_1: from $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i) \cdots] = 0$,

$$
\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_2\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_2 x_1)\partial_{x_1} + (\mathcal{O}_2 x_2)\partial_{x_2}\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right]. \qquad (38)
\end{aligned}
$$

$$
\begin{aligned}
0 =& D R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] - b_2 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_1 \Big(R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] + R_1[\{n_3, n_4\}, \{a\}, \{b_1 + 2, b_2 - 2\}] \\
& - k_s^2 R_1[\{n_3, n_4\}, \{a\}, \{b_1 + 2, b_2\}] \Big) \\
& + R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 - 1\}]. \qquad (39)
\end{aligned}
$$

Equation of motion for R_1:

$$
\begin{aligned}
R_1[\{2, n_4\}, \{a\}, \{b_1, b_2\}] =& - R_1[\{0, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - (2\nu + 1) R_1[\{1, n_4\}, \{a - 1\}, \{b_1, b_2 + 1\}], \qquad (40) \\
R_1[\{n_3, 2\}, \{a\}, \{b_1, b_2\}] =& - R_1[\{n_3, 0\}, \{a\}, \{b_1, b_2\}] \\
& - (2\nu + 1) R_1[\{n_3, 1\}, \{a - 1\}, \{b_1, b_2 + 1\}]. \qquad (41)
\end{aligned}
$$

Symmetry relations: R_1 inherits symmetry from the bubble:

$$
R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = R_1[\{n_4, n_3\}, \{a\}, \{b_1, b_2\}]. \qquad (42)
$$

More importantly, R_1 and R_2 are related by loop-momentum shift $\boldsymbol{q} \to -(\boldsymbol{q} + \boldsymbol{k}_s)$:

$$R_2[\{n_1, n_2\}, \{a\}, \{b_1, b_2\}] = R_1[\{n_1, n_2\}, \{a\}, \{b_2, b_1\}]. \tag{43}$$

This symmetry implies that one only needs to implement the IBP system for R_1; all relations for R_2 follow by applying $b_1 \leftrightarrow b_2$ and appropriate index relabeling.

These tadpole integrals are simpler than the double-time G integrals because they involve only one time variable τ and two Hankel functions (vs four in G), and can be reduced to master integrals using the above IBP system.

Differential operators for R_1 From the definition of R_1, we can derive relations by taking derivatives with respect to parameters:

Derivative with respect to P_0:

$$\frac{\partial}{\partial P_0} R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = i R_1[\{n_3, n_4\}, \{a+1\}, \{b_1, b_2\}]. \tag{44}$$

Derivative with respect to k_s:

$$\begin{aligned}
&\frac{\partial}{\partial k_s} R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
&= -\frac{b_2}{2k_s}\Big(k_s^2\, R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2+2\}] \\
&\qquad + R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
&\qquad - R_1[\{n_3, n_4\}, \{a\}, \{b_1-2, b_2+2\}]\Big) \\
&+ \frac{1}{2k_s}\Big(k_s^2\, R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1, b_2+1\}] \\
&\qquad + R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1, b_2-1\}] \\
&\qquad - R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1-2, b_2+1\}]\Big) \\
&+ \frac{1}{2k_s}\Big(k_s^2\, R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1, b_2+1\}] \\
&\qquad + R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1, b_2-1\}] \\
&\qquad - R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1-2, b_2+1\}]\Big).
\end{aligned} \tag{45}$$

Scaling relation (dimensional analysis):

$$\left(k_s \frac{\partial}{\partial k_s} + P_0 \frac{\partial}{\partial P_0}\right) R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = (D - b_1 - b_2 - a - 1) R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}]. \tag{46}$$

This scaling relation will be used to verify the correctness of differential equations.

3.3 Odevity seclection rules for the IBP system

There are two useful parity classifications ("Odevity") for the bubble IBP system:

1. For dq-IBP and dτ-IBP, the parities of $n_1 + n_2 + b_1$ and $n_3 + n_4 + b_2$ are conserved.

2. For dq-IBP, the parities of $n_1 + n_3 + a_1$ and $n_2 + n_4 + a_2$ are conserved.

In reduction runs we restrict the IBP system to the even sector of the first classification, i.e.

$$n_1 + n_2 + b_1 \text{ even}, \qquad n_3 + n_4 + b_2 \text{ even}. \tag{47}$$

For the tadpole families this implies two independent constraints:

- For R_1 (obtained by contracting indices where one is 1 and the other is 0), one must impose b_1 even *and* $n_3+n_4+b_2$ even. This contraction produces a term with an odd power (excluding the cancelling 2ν shift), which matches the Odevity of the top sector.

- For R_2, the loop-momentum shift symmetry (43) exchanges $b_1 \leftrightarrow b_2$, so the corresponding restriction is b_2 even and $n_1 + n_2 + b_1$ even.

This is why Odevity belongs to the IBP chapter: it specifies which integrals are allowed to appear in the closed IBP system that we export to Kira.

3.4 Reduction strategy

1. Select a Odevity sub-system of IBP-system. Apply EoM and symmetry to time-IBP and momentum-IBP, Then feed the processed IBP relations to Kira for reduction, since Mathematica is too slow for the reduction step.

2. Reduce R_1, R_2 first. then, reduce G with the known R_1, R_2 in time IBP.

4 Baikov representation and dimension-shift relations

Baikov variables for the one-loop two-point momentum integral Introduce $x_1 \equiv |\boldsymbol{q}|$, $x_2 \equiv |\boldsymbol{q} + \boldsymbol{k}_s|$ and

$$z_1 = x_1^2 = q^2, \qquad z_2 = x_2^2 = (q + k_s)^2, \qquad \mathcal{K} = k_s^2, \tag{48}$$

with Gram determinant

$$\mathcal{G} = \frac{1}{4}\Big(2z_1k_s^2 + 2z_2k_s^2 - k_s^4 - z_1^2 - z_2^2 + 2z_1z_2\Big) = \frac{1}{4}\lambda(z_1, z_2, k_s^2), \tag{49}$$

where λ is the Källén function. The one-loop Baikov change of variables gives (for general d)

$$\mathrm{d}^d q \to C_d\, \mathcal{G}^{\frac{d-3}{2}} \mathcal{K}^{-\frac{d-2}{2}}\, \mathrm{d}z_1\, \mathrm{d}z_2, \qquad C_d = \frac{\pi^{-1/2}}{\Gamma[(d-1)/2]}. \tag{50}$$

In particular, since $\mathrm{d}z_i = 2x_i\, \mathrm{d}x_i$, one has the useful bookkeeping relation

$$G[\{\boldsymbol{n}\}, \{a_1, a_2\}, \{b_1 + 1, b_2 + 1\}] \sim \cdots \times \frac{\mathrm{d}z_1}{x_1^{b_0+b_1+1}} \frac{\mathrm{d}z_2}{x_2^{b_0+b_2+1}} = \cdots \times 4\, \frac{\mathrm{d}x_1}{x_1^{b_0+b_1}} \frac{\mathrm{d}x_2}{x_2^{b_0+b_2}}. \tag{51}$$

The integration region is determined by $\mathcal{G} \geq 0$, equivalently the triangle inequalities between (x_1, x_2, k_s).

Dimension shift With generalized Sylvester's determinant identity, one has

$$|\partial_{x_{ij}}|_{i,j\le n_1}|x_{ij}|^\gamma = |x_{ij}|_{i,j>n_1}\left(\prod_{i=0}^{n_1-1}(\gamma+i)\right)|x_{ij}|^{\gamma-1}. \tag{52}$$

For a symmetric matrix $x_{ij}=x_{ji}$ one further has

$$\begin{aligned}&|a_{ij}\partial_{x_{ij}}|_{i,j\le n_1}|x_{ij}|^\gamma = |x_{ij}|_{i,j>n_1}\left(\prod_{i=0}^{n_1-1}\left(\gamma+\frac{i}{2}\right)\right)|x_{ij}|^{\gamma-1}\\ &a_{ij}=\frac{1}{2}\ \text{ for }\ i\neq j,\quad a_{ij}=1\ \text{ for }\ i=j.\end{aligned} \tag{53}$$

Now take

$$x_{ij}=q_i\cdot q_j,\qquad n_1=L,\qquad q_i=l_1,\dots,l_L,p_1,\dots,p_E, \tag{54}$$

where l_i are loop momenta and p_i are external momenta. Then $|x_{ij}|=\mathcal{G}$ and $|x_{ij}|_{i,j>n_1}=\mathcal{K}$ are Gram determinants. Writing

$$Q=\frac{1}{\prod_i z_i^{a_i}},\qquad \gamma=\frac{d-E-L-1}{2}, \tag{55}$$

the identity above leads to a dimension-raising relation in Baikov form:

$$\begin{aligned}C_{\gamma-1}\int Q\,\frac{\mathcal{G}^{\gamma-1}}{\mathcal{K}^{\gamma-1+L/2}}\prod \mathrm{d}z_i &= C_\gamma\int Q\,|a_{ij}\partial_{x_{ij}}|_{i,j\le L}\,\frac{\mathcal{G}^\gamma}{\mathcal{K}^{\gamma+L/2}}\prod \mathrm{d}z_i\\ &= (-1)^L C_\gamma\int Q\,|\overleftarrow{a_{ij}\partial_{x_{ij}}}|_{i,j\le L}\,\frac{\mathcal{G}^\gamma}{\mathcal{K}^{\gamma+L/2}}\prod \mathrm{d}z_i,\end{aligned} \tag{56}$$

where the left-hand side corresponds to a $d-2$ dimensional integral while the right-hand side produces a d dimensional one after integrating by parts.

5 Intersection theory viewpoint and dlog integrands

By intersection theory, if a master-integrand basis ϕ_i is of $d\log$ form and the twist u satisfies $\mathrm{d}u=(\mathrm{d}\log u)\,u$, then the differential equations for the master integrals $\int\phi_i u$ take a $d\log$ form. Here we use a mild generalization motivated by fibration intersection theory: treat the master objects as $\Phi_i\cdot U$ with both Φ_i and U viewed as vectors, choose U to satisfy

$$\mathrm{d}U=\mathrm{d}\Omega\cdot U, \tag{57}$$

and construct Φ_i as $d\log$-form integrands. This motivates a canonical-like choice of masters for the de Sitter bubble.

Time-integral block as a first-order system Define the "tree-level-like" time integral (no momentum denominators)

$$I_{\boldsymbol{n}}(x_1,x_2;P_1,P_2) \equiv \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2\, e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\,(-\tau_1)^{a_0}(-\tau_2)^{a_0} \times h^{(1)}_{n_1}(-x_1\tau_1)h^{(2)}_{n_2}(-x_1\tau_2)\,h^{(1)}_{n_3}(-x_2\tau_1)h^{(2)}_{n_4}(-x_2\tau_2). \tag{58}$$

For fixed (ν, a_0), the $I_{\boldsymbol{n}}$ span a finite basis and satisfy a first-order system

$$\mathrm{d}I_{\boldsymbol{m}} = (\mathrm{d}\Omega_{\boldsymbol{mn}})\, I_{\boldsymbol{n}}, \tag{59}$$

where the one-forms in $\mathrm{d}\Omega$ are built from $\mathrm{d}\log x_1$, $\mathrm{d}\log x_2$ and $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$. For the two-vertex tree-level-like family entering the bubble, there are 2^4 time-integral master functions.

$$\mathrm{d}\Omega = \mathrm{d}\Omega_{x_1} + \mathrm{d}\Omega_{x_2} + \mathrm{d}\Omega_{ex}, \tag{60}$$

where $\mathrm{d}\Omega_{x_i}$ only contains $\mathrm{d}\log x_i$ while $\mathrm{d}\Omega_{ex}$ is composed of $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$ with constant coefficients (built from ν and a_0).

Baikov twist and dlog building blocks With ϵ the dimensional regulator ($d = 3 - 2\epsilon$), the one-loop Baikov measure contributes the factor $\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}$, and one can package the combined twist schematically as a vector

$$U = x_1^{-b_0} x_2^{-b_0}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\, I_{\boldsymbol{n}}, \tag{61}$$

whose differential is again a sum of $d\log$ one-forms,

$$\mathrm{d}U = \Big(\mathrm{d}\Omega - b_0(\mathrm{d}\log x_1 + \mathrm{d}\log x_2) - \epsilon(\mathrm{d}\log\mathcal{G} - \mathrm{d}\log\mathcal{K})\Big)\cdot U. \tag{62}$$

In terms of $z_i = x_i^2$, useful $d\log$ building blocks include

$$\begin{aligned} &\mathrm{d}\log z_1 = 2\mathrm{d}\log x_1, \qquad \mathrm{d}\log z_2 = 2\mathrm{d}\log x_2, \\ &\frac{\sqrt{z_2 k_s^2}\,\mathrm{d}z_1}{\mathcal{G}} = -\mathrm{d}\log\left(\frac{2\sqrt{z_2k_s^2}+k_s^2-z_1+z_2}{-2\sqrt{z_2k_s^2}+k_s^2-z_1+z_2}\right), \qquad z_1 \leftrightarrow z_2. \end{aligned} \tag{63}$$

Since $\mathrm{d}\int \mathrm{d}x_i/(P_i \pm x_1 \pm x_2)$ is of $d\log$ form, one also has the building blocks

$$\begin{aligned} \partial_{P_j}\Omega_{ex}\,\mathrm{d}x_i &= \partial_{P_j}\Omega\,\mathrm{d}x_i, \\ \partial_{x_j}\Omega_{ex}\,\mathrm{d}x_i &= (\partial_{x_j}\Omega - \partial_{x_j}\Omega_{x_j})\,\mathrm{d}x_i. \end{aligned} \tag{64}$$

Using

$$\begin{aligned} G[\{\boldsymbol{n}\},\{a_1+1,a_2\},\{b_1,b_2\}] &= \partial_{P_1}G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] = (\partial_{P_1}\Omega_{\boldsymbol{nm}}) \\ &\quad\times G[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1,b_2\}], \\ G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] &= \frac{1}{b_0+b_1-1}\,(\partial_{x_1}\Omega_{\boldsymbol{nm}}) \\ &\quad\times G[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1-1,b_2\}], \end{aligned} \tag{65}$$

one may view $\partial_{P_j}\Omega\,\mathrm{d}x_i \sim \tau_j\,\mathrm{d}x_i$ and $\partial_{x_j}\Omega\,\mathrm{d}x_i \sim \frac{b_0+b_j-1}{x_j}\mathrm{d}x_j$ as additional $d\log$-type building blocks at the integrand level.

5.1 Top-sector $d\log$ candidates

Constructing candidate $d\log$ integrands For notational convenience, we split off common shifts and write

$$G[\{\boldsymbol{n}\},\{a_0+a_1,a_0+a_2\},\{b_0+b_1,b_0+b_2\}] \to G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}], \tag{66}$$

where $a_i, b_i \in \mathbb{Z}$ and (a_0, b_0) encode the universal shifts induced by the special-function normalization and prefactors.

Some candidate $d\log$-type integrands in $d = 3-2\epsilon$ are (up to overall constants)

$$\begin{aligned}
&k_s\, G[\{\boldsymbol{n}\},\{0,0\},\{2,2\};d=3] \sim \frac{\mathrm{d}x_1\,\mathrm{d}x_2}{x_1x_2}\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
&k_s\, G[\{\boldsymbol{n}\},\{1,0\},\{2,1\};d=3] \sim \frac{\mathrm{d}x_1}{x_1}\,\tau_1\,\mathrm{d}x_2\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
&k_s\, G[\{\boldsymbol{n}\},\{1,1\},\{1,1\};d=3] \sim \tau_1\,\mathrm{d}x_1\,\tau_2\,\mathrm{d}x_2\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \qquad \text{[CJQ: wrong, double pole]}.
\end{aligned} \tag{67}$$

In $d = 1-2\epsilon$, one can also use the Baikov building block $\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}$ to form $d\log$ candidates such as

$$\begin{aligned}
G[\{\boldsymbol{n}\},\{0,0\},\{1,0\};d=1] &\sim \frac{\mathrm{d}z_1}{z_1}\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
G[\{\boldsymbol{n}\},\{1,0\},\{0,0\};d=1] &\sim \tau_1\,\mathrm{d}x_1\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned} \tag{68}$$

In addition, one may also consider candidates generated by Ω_{ex}:

$$\begin{aligned}
&(\partial_{x_2}\Omega_{ex}\,\mathrm{d}x_1)\,\frac{\mathrm{d}x_2}{x_2}\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
&(\partial_{x_2}\Omega_{ex}\,\mathrm{d}x_1)\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned} \tag{69}$$

5.2 Tadpole-family $d\log$ candidates

Tadpole families R_1, R_2 and $d\log$ candidates The tadpole terms R_1, R_2 can be treated similarly at the integrand level, by isolating the one-time "tree-level-like" block

$$I^{(R)}_{n_3n_4}(x_2;P_0) \equiv \int_{-\infty}^{0} \mathrm{d}\tau\, e^{-\mathrm{i}P_0\tau}\,(-\tau)^{a_0}\, h^{(1)}_{n_3}(-x_2\tau)h^{(2)}_{n_4}(-x_2\tau). \tag{70}$$

For fixed (ν, a_0), the $I^{(R)}_{n_3n_4}$ form a finite basis and satisfy a first-order system

$$\mathrm{d}I^{(R)}_{\boldsymbol{m}} = \left(\mathrm{d}\Omega^{(R)}_{\boldsymbol{mn}}\right)\, I^{(R)}_{\boldsymbol{n}}, \tag{71}$$

where $\mathrm{d}\Omega^{(R)}$ is built from $\mathrm{d}\log x_2$ and $\mathrm{d}\log(P_0 \pm x_2)$. The combined twist can be packaged as

$$U^{(R)} = x_1^{-b_0}x_2^{-b_0}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\, I^{(R)}_{\boldsymbol{n}}. \tag{72}$$

Some candidate $d\log$-type integrands for R_1 in $d=3-2\epsilon$ are (up to overall constants)

$$
\begin{aligned}
k_s\, R_1[\{n_3,n_4\},\{0\},\{2,2\};d=3] &\sim \frac{\mathrm{d}x_1\,\mathrm{d}x_2}{x_1x_2}\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
k_s\, R_1[\{n_3,n_4\},\{1\},\{2,1\};d=3] &\sim \frac{\mathrm{d}x_1}{x_1}\,\tau\,\mathrm{d}x_2\, x_1^{-b_0}x_2^{-b_0}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned}
\tag{73}
$$

In $d=1-2\epsilon$, one can also use the Baikov building blocks $\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}$ and $\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}$ to form additional $d\log$ candidates such as

$$
\begin{aligned}
R_1[\{n_3,n_4\},\{0\},\{1,0\};d=1] &\sim \frac{\mathrm{d}z_1}{z_1}\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
R_1[\{n_3,n_4\},\{0\},\{0,1\};d=1] &\sim \frac{\mathrm{d}z_2}{z_2}\,\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
R_1[\{n_3,n_4\},\{1\},\{0,0\};d=1] &\sim \tau\,\mathrm{d}x_1\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
R_1[\{n_3,n_4\},\{1\},\{0,0\};d=1] &\sim \tau\,\mathrm{d}x_2\,\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}\, z_1^{-b_0/2}z_2^{-b_0/2}\, I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned}
\tag{74}
$$

Raising $d=1$ back to $d=3$ For the Baikov measure one has

$$
\mathrm{d}^d q \to C_d\,\mathcal{G}^{\frac{d-3}{2}}\mathcal{K}^{-\frac{d-2}{2}}\,\mathrm{d}z_1\,\mathrm{d}z_2. \tag{75}
$$

In particular, for $d=1-2\epsilon$ and $d=3-2\epsilon$,

$$
C_{1-2\epsilon}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} = \partial_{q^2}\left[C_{3-2\epsilon}\,\mathcal{K}^{-1/2+\epsilon}\mathcal{G}^{-\epsilon}\right], \tag{76}
$$

where $\partial_{q^2}\equiv\partial_{z_1}+\partial_{z_2}$ and we used $\partial_{q^2}\mathcal{G}=\mathcal{K}$ together with $\Gamma(1-\epsilon)=-\epsilon\,\Gamma(-\epsilon)$.

Therefore, for

$$
\begin{aligned}
G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] = \mathrm{d}z_1\,\mathrm{d}z_2\, z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2}\, I_{\boldsymbol{n}}\\
\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\, C_{1-2\epsilon}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2},
\end{aligned}
\tag{77}
$$

integration by parts gives

$$
\begin{aligned}
G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] = \mathrm{d}z_1\,\mathrm{d}z_2\left[\left(\frac{b_0+b_1}{2z_1}+\frac{b_0+b_2}{2z_2}\right)I_{\boldsymbol{n}} - (\partial_{z_1}+\partial_{z_2})\,I_{\boldsymbol{n}}\right]\\
\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\, z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2}\, C_{3-2\epsilon}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.
\end{aligned}
\tag{78}
$$

Equivalently, in (x_1,x_2) variables with $\partial_{q^2}=\frac{1}{2x_1}\partial_{x_1}+\frac{1}{2x_2}\partial_{x_2}$,

$$
\begin{aligned}
G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] = 2\,\mathrm{d}x_1\,\mathrm{d}x_2\left[\left(\frac{b_0+b_1}{x_1^2}+\frac{b_0+b_2}{x_2^2}\right)I_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right)I_{\boldsymbol{n}}\right]\\
\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}\, x_1^{-(b_0+b_1-1)}x_2^{-(b_0+b_2-1)}\, C_{3-2\epsilon}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.
\end{aligned}
\tag{79}
$$

In particular, for the $d=1$ candidates above,

$$
\begin{aligned}
G[\{\boldsymbol{n}\},\{0,0\},\{1,0\};d=1] &\sim \mathrm{d}z_1\,\mathrm{d}z_2\,z_1^{-b_0/2-1/2}z_2^{-b_0/2}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2}\\
&= 2\,\mathrm{d}x_1\,\mathrm{d}x_2\left[\left(\frac{b_0+1}{x_1^2}+\frac{b_0}{x_2^2}\right)I_{\boldsymbol{n}}-\left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right)I_{\boldsymbol{n}}\right]\\
&\quad\times x_1^{-b_0}x_2^{-b_0+1}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon},\\
G[\{\boldsymbol{n}\},\{1,0\},\{0,0\};d=1] &= 2\,\mathrm{d}x_1\,\mathrm{d}x_2\left[b_0\left(\frac{1}{x_1^2}+\frac{1}{x_2^2}\right)I_{\boldsymbol{n}}-\left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right)I_{\boldsymbol{n}}\right]\\
&\quad\times(-\tau_1)\,x_1^{-b_0+1}x_2^{-b_0+1}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.
\end{aligned}
\tag{80}
$$

Moreover,

$$
\begin{aligned}
&(\partial_{x_2}\Omega_{ex}\,\mathrm{d}x_1)\,\frac{\mathrm{d}x_2}{x_2}\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=(\partial_{x_2}\Omega-\partial_{x_2}\Omega_{x_2})\,\mathrm{d}x_1\,\frac{\mathrm{d}x_2}{x_2}\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=\mathrm{d}x_1\,\mathrm{d}x_2\,(\partial_{x_2}I_{\boldsymbol{n}})\,x_1^{-b_0}x_2^{-b_0-1}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}-\mathrm{d}x_1\,\mathrm{d}x_2\,(\partial_{x_2}\Omega_{x_2})\,x_1^{-b_0}x_2^{-b_0-1}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=\mathrm{d}x_1\,(\mathrm{d}I_{\boldsymbol{n}})\,x_1^{-b_0}x_2^{-b_0-1}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}-\mathrm{d}x_1\,\mathrm{d}x_2\,(\partial_{x_2}\Omega_{x_2})\,x_1^{-b_0}x_2^{-b_0-1}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=-\mathrm{d}x_1\,I_{\boldsymbol{n}}\,\mathrm{d}\big(x_1^{-b_0}x_2^{-b_0-1}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\big)-\mathrm{d}x_1\,\mathrm{d}x_2\,(\partial_{x_2}\Omega_{x_2})\,x_1^{-b_0}x_2^{-b_0-1}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=\left(\frac{b_0+1}{x_2}-\partial_{x_2}\Omega_{x_2}\right)k_s\,x_1^{-b_0}x_2^{-b_0-1}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon-1/2}\,\mathrm{d}x_1\,\mathrm{d}x_2.
\end{aligned}
$$

and similarly

$$
\begin{aligned}
&(\partial_{x_2}\Omega_{ex}\,\mathrm{d}x_1)\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=\left(\frac{b_0-1}{x_2}-\partial_{x_2}\Omega_{x_2}\right)x_1^{-b_0+1}x_2^{-b_0+1}\,I_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2}\,\mathrm{d}x_1\,\mathrm{d}x_2.
\end{aligned}
$$

The practical goal is to pick a master basis for the full de Sitter bubble (including the special-function time dependence) such that the induced differential system in external kinematics is as close as possible to a $d\log$ form.

6 Differential equations for the tadpole master integrals

6.1 Master-integral definitions

The Kira implementation uses shifted indices

$$
(-\tau)^{a_0+a},\qquad |q|^{-b_1},\qquad |\boldsymbol{q}+\boldsymbol{k}_s|^{-b_2},
\tag{81}
$$

with

$$
a_0=2\delta_a+2\nu,\qquad b_{10}=0,\qquad b_{20}=-2\nu,\qquad D=3-2\epsilon.
\tag{82}
$$

In this convention the tadpole family is

$$R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] \equiv e^{\pi \mathrm{Im}\,[\nu]} \frac{4i}{\pi} \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_0\tau} \times \frac{(-\tau)^{a_0+a}}{|q|^{b_{10}+b_1}\,|\boldsymbol{q}+\boldsymbol{k}_s|^{b_{20}+b_2}} h^{(1)}_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau) h^{(2)}_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau). \tag{83}$$